MILITARY STANDARDIZATION HANDBOOK

RUBBER PRODUCTS: RECOMMENDED SHELF LIFE

TO ALL HOLDERS OF MIL-HDBK-695C

1. THE FOLLOWING PAGES OF MIL-HDBK-695C HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
5	14 November 1986	5	27 March 1985
6	27 March 1985	6	REPRINTED WITHOUT CHANGE
7	14 November 1986	7	27 March 1985
8	27 March 1985	8	REPRINTED WITHOUT CHANGE
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2. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

3. Holders of MIL-HDBK-695C will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the military handbook is completely revised or cancelled.

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Review activities:
Army - GL, MI, ME, AR
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TABLE I. Age resistance generally associated with products fabricated from various rubbers.

Type of Rubber	Common or Trade Name	ASTM D1418 Abbreviation	ASTM D2000 Abbreviation	MIL-STD-417 Designation
		Up to 20 Years		
Silicone	Silicone	Q	FE	TA
Fluorosilicone	Silastic LS	FVMQ	FK	TA
Polysulfide	Thiokol	T	BK	SA
Fluorocarbons	Fluorel, Viton	FKM	HK	-
Polyacrylate	Acrylic	ACM, ANM	DF, DH	TB
Fluorophosphonitrilic	[1]PNF, Eypel-F	FZ	-	TB
Perfluorocarbon	KALREZ	FFKM	-	-
		5 to 10 Years		
[2]Butadiene/ acrylonitrile	Nitrile, NBR	NBR	BF, BG, CH	SB
Chlorosulfonated polyethylene	Hypalon	CSM	CE	-
Isobutylene/ Isoprene	Butyl	IIR	AA, BA	RS
Polychloroprene	Neoprene	CR	BC, BE	SC
Polyether Urethane	Urethane	EU	BG	-
Polypropylene oxide	Propylene oxide	GPO	-	-
Ethylene/propylene diene	Ethylene propylene terpolymer	EPDM	BA, CA	-
Ethylene/propylene	Ethylene propylene copolymer	EPM	BA, CA	-
Epichlorohydrin homopolymer	Hydrin 100, Herclor H	CO	-	-
Epichlorohydrin ethylene oxide	Hydrin 200, Herclor C	ECO	-	-
		3 to 5 Years		
[2]Butadiene/ acrylonitrile	Nitrile, NBR	NBR	BF, BG, CH	SB
Butadiene/styrene	SBR	SBR	AA, BA	RS
Cis-polybutadiene	Butadiene	BR	AA	RN
Cis 1, 4, poly-isoprene	Natural, pale crepe	NR	AA	RN
Cis 1, 4, poly-isoprene	Synthetic natural	IR	AA	RN
[3]Polyester urethane	Urethane	AU	-	-

1. Common or Trade Name "PNF" no longer being produced.
2. See table II through VIII.
3. Some polyester urethanes have very poor resistance to water and humidity.

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TABLE II. Military specifications.

Number	Title	Elastomers[1] Required	Elastomers[2] Commonly Used	Expected Shelf Storage Life, Years
MIL-C-17	Cables, Radiofrequency; Coaxial, Dual Coaxial, Twin Conductor, & Twin Lead			
	Core Dielectric:			
	Types D, E and H	-	-	3
	Type J	IIR	-	10
	Type L	Q	-	20
	Jacket:			
	Type IV	CR	-	8
	Type VI	Q	-	20
	Type VIII	CR	-	8
MIL-H-370	Hose and Hose Assemblies, Non-Metallic; Rubber Liquid Fuel, Non Collapsible			
	Types 1 and 2	NBR (Tube)	-	3
MIL-W-432	Gasket Nonmetallic, Synthetic Rubber	-	NBR	3
MIL-C-442	Cable, (Wire), Two Conductor, Parallel (Ripcord)			
	Types IV, V and VI	SBR	-	3
MIL-L-741	Leads, Electrical, Arc Welding	SBR	-	3
MIL-O-836	Overshoe, Rubber (Man's, 5 Buckle Type)			
	Type I and II	NR or SYN[3]	-	3
MIL-C-882	Cloth, (Cotton Duck), Laminated, Synthetic Rubber Impregnated, Oil Resistant			
	Class I	CR	-	5
	Class II	NBR	-	3
MIL-R-900	Rubber Gasket Material, 45 Durometer Hardness	-	SBR	3

NOTES:

[1]If no elastomer is listed, none is required by the specification.

[2]If no elastomer is listed, the elastomer commonly used is unknown.

[3]SYN = synthetic (for other abbreviations see Table I).

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TABLE II. Military specifications (cont'd).

Number	Title	Elastomers[1] Required	Elastomers[2] Commonly Used	Expected Shelf Storage Life, Years
MIL-M-910	Mats, Floor, Standing	-	-	3
MIL-C-915	Cable and Cord, Electrical for Shipboard Use, (Gen. Spec. for)			
	Type CVSF	SYN & CR	-	5
	Type DCOP	SYN & CR	-	5
	Type DHOF	IIR & CR	-	8
	Type DSS	SYN & CR	-	5
	Type DLT	NR & CR	-	5
	Type FSCF	IIR & CR	-	8
	Type FHOF	IIR & CR	-	8
	Type FSS	SYN & CR	-	5
	Type JAS	IIR & CR	-	8
	Type MCSC	SYN & CR	-	5
	Type MCOS	CR	-	8
	Type MSS	SYN & CR	-	5
	Type SDU	IIR & CR	-	8
	Type SSF	SYN & CR	-	5
	Type TCOP	SYN & CR	-	5
	Type THOF	IIR & CR	-	8
	Type TRF	SYN & CR	-	5
	Type TRXF	CR	-	8
	Type TSS	SYN & CR	-	5
MIL-G-1086	Gasket Material, Synthetic Rubber (For Bolted Steel & Aluminum Tanks)			
	Type I	-	CR	8
	Type II	-	NBR	3
MIL-G-1149	Gasket Material Synthetic Rubber, 50 and 65 Durometer Hardness			
	Class 1 - Types I & II	CR	-	8
	Class 2 - Types I & II	SBR	-	3
	Class 3 - Types I & II	IIR	-	10
	Class 5 - Types I & II	NBR	-	3
MIL-R-1229	Rubber Sheet Packing, Solid, Wire Cloth Inserted Symbol 2354	-	SBR	3

NOTES:

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TABLE II. Military specifications (cont'd).

Number	Title	Elastomers[1] Required	Elastomers[2] Commonly Used	Expected Shelf Storage Life, Years
MIL-R-1632	Rubber Band, Parachute Suspension Line Type I	NR	-	3
MIL-T-1956	Tarpaulin, Waterproof, Special Purpose, 10 Feet Long by 8 Feet Wide All Classes	-	-	3
MIL-H-2217	Hose Assemblies, Wire-Reinforced Rubber, Submarine Rescue Chamber	NR (Tube)	-	3
MIL-C-2486	Cable, Special Purpose, Electrical WM-46/U	SBR	-	3
MIL-P-2693	Packing Material, Cold Storage Door Gasketing, Nonwatertight			
	Type A	-	CR	8
	Type B	-	NR (core)	2
MIL-R-2765	Rubber Sheet, Strip, Extruded, & Molded Shapes, Synthetic, Oil Resistant	-	NBR	3
MIL-H-2815	Hose Assemblies, Rubber, Diver's Breathing Air and Gas Supply	NR (Tube)	-	3
MIL-J-2829	Joint Sealing Packing			
	Types I & II - Classes 1 & 2	-	CR	8
MIL-B-2885	Boots, Firemen's	NR or SYN	-	3
MIL-P-2911	Packing Assembly, Hydraulic, Conical & V Types	-	-	3
MIL-S-2912	Synthetic Rubber Compound, Acid & Oil Resistant (For Lining Battery Compartments on Submarines)			
	Class 1 - Types I & II	CR	-	5
	Class 2 - Types I & II	NBR	-	3
MIL-D-2921	Disks, Rubber (Cellular, Hard)			
	Type A	Reclaim	-	3
	Type B - Class a	SBR	-	3
	Type B - Class b	NBR	-	3
MIL-G-3036	Grommets - Elastic, Hot-Oil & Coolant-Resistant	-	NBR	3
MIL-R-3065	Rubber, Fabricated Products (see MIL-STD-417)			

NOTES:
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[2] If no elastomer is listed, the elastomer commonly used is unknown.

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TABLE II. Military specifications (cont'd).

Number	Title	Elastomers[1] Required	Elastomers[2] Commonly Used	Expected Shelf Storage Life Years
MIL-B-50015	Bulb, Aspirator	NR	-	3
MIL-F-50070	Faceblank, C17	NR	-	3
MIL-H-51059	Hose, Fuel, Flame Thrower, M8	NBR (Tube)	-	5
MIL-F-51109	Faceblank, Protective Mask C12R4	NR	-	3
MIL-H-52079	Hose, Preformed: Semi-Flexible, Reinforced	-	-	3
MIL-H-52262	Hose Assembly, Rubber, Lightweight, Collapsible, 4-inch	NBR (Tube) SYN (Cover)	-	3
MIL-C-52286	Cable & Assemblies, Power, Electrical (with Molded on Terminations)	-		
MIL-H-52471	Hose & Hose Assemblies, Rubber, Hydraulic Pressure Type (Gen. Spec. for)	NBR (Tube) CR (Cover)	-	3
MIL-H-52544	Hose Rubber, Oil Suction, Wire Reinforced	CR (Tube) NBR (Cover)	-	3
MIL-B-52761	Belting, Flat: Conveyor, Rubber Industrial Synthetic Reinforcement	NR-SYN	-	3
MIL-C-55036	Cable, Telephone, WM-130 ()/G	CR (Jacket)	-	5
MIL-C-55040	Cable, Special Purpose, Electrical (Retractile)	-	-	5
MIL-C-55479[3]	Cable, Assembly CX-162 & CX-163	-	CR	8
MIL-C-55480	Cable, Telephone WC-534	-	CR	8
MIL-C-55483	Cable Power Electrical Flat Three Conductor	CR	-	8
MIL-T-62118	Tread, Rubber, Strip Form Stock For Retreading Pneumatic Tires	NR, IR, SBR, BR	-	3
MIL-T-62129	Tire, Pneumatic, For Truck, Logistical Type	-	-	3
MIL-W-62136	Wheel & Tire Assemblies, Foam Inflated Low Speed Highway & Off-Road	-	-	3
MIL-T-62157	Tire, Pneumatic, With Flap, 1400-20 Run Flat	-	-	3
MIL-R-810903	Rubber, Electrically Conductive Fuel Resistant	-	NBR	3
MIL-H-81581	Hose Assembly, Breathing Oxygen & Air	Q	-	20
MIL-P-81716	Packing, Preformed, Straight Thread Tube Fitting Boss, Type II Hydraulic (Minus 65 F to Plus 275 F)	-	-	10

NOTES:
[1]If no elastomer is listed, none is required by the specification.
[2]If no elastomer is listed, the elastomer commonly used is unknown.
[3]Cancelled, no replacement.

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TABLE II. Military specifications (cont'd).

Number	Title	Elastomers[1] Required	Elastomers[2] Commonly Used	Expected Shelf Storage Life Years
MIL-R-81628	Rubber, Chlorosulfonated Polyethylene Elastomer, Sheet & Molded Shapes, Ozone Resistant	CSM	-	10
MIL-H-82127	Hose Assembly, Rubber (Synthetic): Fuel, Discharge, Collapsible	SYN	-	3
MIL-H-82142	Hood, Rain, Woman's: Nylon, Rubber Coated Green	NBR	-	3
MIL-R-82190	Raincoat, Woman's, Nylon, Rubber Coated Green	NBR	-	3
MIL-G-82242	Glove Shells, Radioactive Contaminants, Protective	NR	-	3
MIL-C-82255	Cloth, Coated, Snythetic Rubber (Nitrile & Polychloroprene)			
	Types I, II, IV	CR	-	5
	Type III	NBR	-	3
MIL-S-82258	Swim Fins, Rubber	NR or IR	-	3
MIL-T-82270	Trousers, Wet Suit (Chloroprene Foam)	CR	-	3
MIL-M-82272	Mittens, Wet Suit (Chloroprene Foam)	CR	-	3
MIL-C-83008[3]	Cloth, Coated, Cotton & Nylon, Polyurethane (Microporus) Coated	AU or EU	-	3
MIL-R-83248	Rubber, Fluorocarbon Elastomer; High Temperature Fluid & Compression Set Resistant	FKM	-	20
MIL-R-83285	Rubber, Ethylene-Propylene, General Purpose	EPDM	-	5
MIL-R-83322	Rubber, Carboxy-Nitroso; Nitrogen Tetroxide (N204) Resistant	SYN	-	5
MIL-R-83397	Rubber, Polyurethane, Castable, Humidity Resistant	AU or EU	-	3
MIL-C-83398	Cloth, Coated, Stretch, Polychloroprene Knitted Nylon	CR	-	5
MIL-R-83412	Rubber, Ethylene-propylene, Hydrazine Resistant	EPDM	-	5
MIL-P-83461	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275 F (135 C)	-	-	10

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[3]Cancelled, no replacement.

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TABLE V. Aerospace Materials Specifications (cont'd).

Number	Title	Elastomers[1] Required	Elastomers[2] Commonly Used	Expected Shelf Storage Life Years
AMS 3248	Synthetic Rubber, Phosphate Ester Resistant Ethylene Propylene Type (55-65)	EPDM	-	8
AMS 3249	Synthetic Rubber, Hydrazine Base Fluid Resistant Ethylene Propylene Type (75-85)	EPDM	-	8
AMS 3260	Rubber, Ethylene-Propylene-Diene, EPDM Type	EPDM	-	8
AMS 3270	Chlopoprene Rubber Sheet, Cotton Fabric Reinforced, Weather Resistant	CR	-	5
AMS 3273	Chlopoprene Rubber Sheet, Nylon Fabric Reinforced, Weather Resistant	CR	-	5
AMS 3274	Nitrile Rubber Sheet, Nylon Fabric Reinforced, Aromatic Fuel Resistant	NBR	-	3
AMS 3301	Silicone Rubber, General Purpose (35-45)	Q	-	20
AMS 3302	Silicone Rubber, General Purpose (45-55)	Q	-	20
AMS 3303	Silicone Rubber, General Purpose (55-65)	Q	-	20
AMS 3304	Silicone Rubber, General Purpose (65-75)	Q	-	20
AMS 3305	Silicone Rubber, General Purpose (75-85)	Q	-	20
AMS 3307	Silicone Rubber, Low Compression Set, Non-Oil Resistant (65-75)	Q	-	20
AMS 3315	Silicone Rubber Sheet, Glass Fabric Reinforced, VMQ Type	Q	-	20
AMS 3320	Silicone Rubber Sheet, Glass Fabric Reinforced, Heat & Weather Resistant (60-80)	Q	-	20
AMS 3325	Silicone Rubber, Fuel & Oil Resistant (55-65)	FVMQ	-	20
AMS 3326	Silicone Rubber, Fuel & Oil Resistant (50-65)	FVMQ	-	20
AMS 3327	Silicone Rubber, High Temperature Fuel & Oil Resistant (70-80)	FVMQ	-	20
AMS 3332	Silicone Rubber, Extreme Low Temperature Resistant (15-30)	Q	-	20
AMS 3334	Silicone Rubber, Extreme Low Temperature Resistant (35-45)	Q	-	20

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TABLE V. Aerospace Materials Specifications (cont'd).

Number	Title	Elastomers[1] Required	Elastomers[2] Commonly Used	Expected Shelf Storage Life Years
AMS 3335	Silicone Rubber, Extreme Low Temperature Resistant (45-55)	Q	-	20
AMS 3336	Silicone Rubber, Extreme Low Temperature	Q	-	20
AMS 3337	Silicone Rubber, High & Extreme Low Temperature Resistant (65-75)	Q	-	20
AMS 3338	Silicone Rubber, Extreme Low Temperature Resistant (75-85)	Q	-	20
AMS 3344[3]	Silicone Rubber - 1800 psi (45-55)	Q	-	20
AMS 3345	Silicone Rubber - 1000 psi (45-55)	Q	-	20
AMS 3346	Silicone Rubber - 1000 psi (55-65)	Q	-	20
AMS 3347	Silicone Rubber, 1200 psi, High Modulus (45-55)	Q	-	20
AMS 3348	Silicone Rubber, 1150 psi, High Resiliency	Q	-	20
AMS 3349	Silicone Rubber, 1100 psi, High Resilience (65-75)	Q	-	20
AMS 3356	Silicone Rubber, Lubricating Oil & Compression Set Resistant Electrical Grade (55-65)	Q	-	20
AMS 3357	Silicone Rubber, Lubricating Oil & Compression Set Resistant (65-75)	Q	-	20
AMS 3386	Hose, Synthetic Rubber, Aircraft Fueling, Textile Reinforced Collapsing	-	NBR	3
AMS 3387	Hose, Nitrile Rubber, Aircraft Fueling, Textile Reinforced, Chloroprene Covered Noncollapsing	NBR	-	3
AMS 3388	Hose, Synthetic Rubber, Aircraft Fueling, Single Wire Braid Reinforced, Noncollapsing	-	NBR	3
AMS 3389	Hose, Synthetic Rubber, Aircraft Fueling Double Wire Braid Reinforced, Noncollapsing	-	NBR	3
AMS 3570	Foam, Flexible Polyurethane Open Cell, Medium Flexiblity	AU, EU	-	3
AMS 7257	Rings, Sealing, Perfluorocarbon, Rubber, High-Temperature-Fluid Resistant (75-85)	FFKM	-	20
AMS 7259	Rings, Sealing, Fluorocarbon Rubber, High-Temperature-Fluid Resistant, Very Low Compression Set (85-95)	FKM	-	20

NOTE:

[1]If no elastomer is listed, none is required by the specification.

[2]If no elastomer is listed, the elastomer commonly used is unknown.

[3]Standard inactive for design purposes.

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